LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	29
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fatigue meter operation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fatigue meter operation.
5. Specific Objectives	15.1.11.2 Trainee will be able to explain operation of fatigue meter.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstrate fatigue meter operation on simulator; observe safety precautions regarding electrical isolation; instructor supervises trainee interaction with controls.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a fatigue meter in an aircraft?
(b) How does the fatigue meter relate to the overall electrical power generation system?
(c) Why is monitoring fatigue critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principle of fatigue meter operation • Input signal acquisition from generators • Signal processing and integration logic • Display unit calibration and readout • Alarm threshold settings and indications	L & Ds	PPT, IB	15 min
(b)	• Step-by-step operational procedure • Interpretation of meter readings • Troubleshooting common display faults • Maintenance checks and calibration steps • Recording data for flight logs	L & Ds	PPT, PDF	20 min
(c)	• Impact of fatigue on power generation • Correlation with other electrical parameters • Limitations of the fatigue meter system • Integration with central warning systems • Future trends in fatigue monitoring	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the sequence of events when a fatigue threshold is exceeded.
(b) Describe how the fatigue meter data assists in maintenance scheduling.
(c) Identify two potential causes for inaccurate fatigue meter readings.

12. Home Assignment and Follow Up	Prepare a brief report comparing the fatigue meter operation principles with the parameter selection methods for flight data recorders covered in LP 30.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Flight data recorders, parameters selected methods of recording.	Refer to the master syllabus reference list	LP 30

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min